
DS 401: Optimization for Data Science

Course Code	DS 401
Course Name	Optimization for Data Science
L-T-P-C	3-0-0-3
Prerequisites	IC111-Linear Algebra, IC110-Engineering Mathematics, DS301- Mathematical Foundations of Data Science I
Intended for	Discipline core for B. Tech. Data Science and Engineering, Discipline elective for B. Tech. Computer Science and Engineering, Electrical Engineering, Free elective for other B. Tech. disciplines
Approval	29th BoA (Mar 2019); 20th Senate (Apr 2019), in the B.Tech Data Science and Engineering curriculum

Course Contents

- **Module I:** Affine sets, convex sets, cone, examples – hyperplanes, halfspaces, polyhedra, simplexes, positive semidefinite cones. Operations that preserve convexity. Separating and supporting hyperplanes. Dual cones. (6 lectures)
- **Module II:** Convex function, first and second order conditions, epigraph, operations that preserve convexity, conjugate function. (6 lectures)
- **Module III:** Convex optimization – linear, quadratic, geometric, conic, semidefinite programming. Formulation of - unconstrained, equality constrained, inequality constrained and both – problems. (7 lectures)
- **Module VI:** Duality – Lagrange dual function, bounds on the optimal value. Lagrange dual problem, weak and strong duality, optimality conditions. (8 lectures)
- **Module V:** Gradient methods – gradient descent, Lipschitz functions, smooth functions, projected gradient descent, Frank-Wolfe algorithm, Chebyhev iterations, conjugate gradient, Nesterov's accelerated gradient descent. Dual gradient ascent, ADAM. (10 lectures)
- **Module VI:** Nonconvex optimization – alternating minimization and expectation maximization algorithms, convex relaxations. (5 lectures)

Text Books

1. Boyd, Stephen, and Lieven Vandenberghe, **Convex optimization**, Cambridge university press, 2004.

References

1. Yurii, Nesterov, **Introductory lectures on convex optimization: a basic course**, Kluwer Academic Publishers, 2004.
2. Luenberger, D. G., and Y. Ye, **Linear and nonlinear programming**, Springer, 2008.
3. Nocedal, Jorge, and Stephen Wright, **Numerical optimization**, Springer Science & Business Media, 2006.